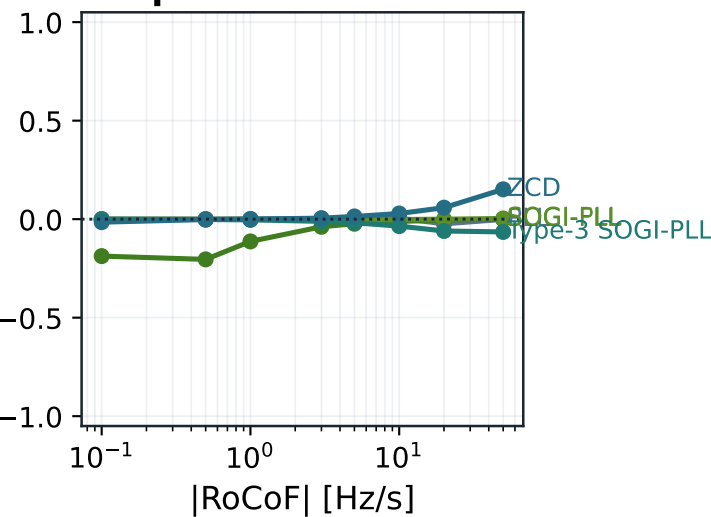


RoCoF sign asymmetry by estimator family

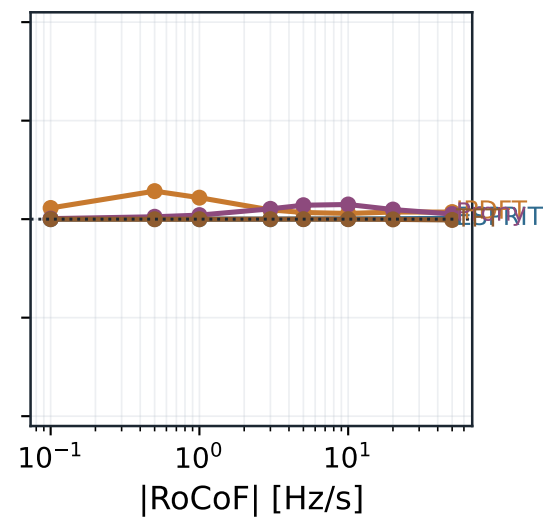
Zero is sign-symmetric. Positive values mean the frequency-decline case is harder than the frequency-rise case.

$\log_{10}(\text{RMSE down-ramp} / \text{RMSE up-ramp})$

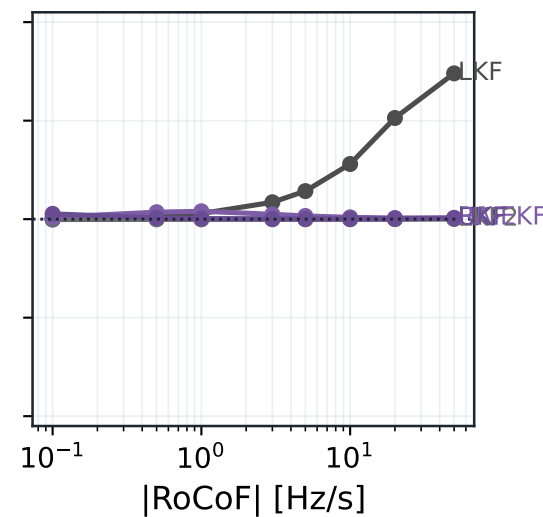
Loop-based



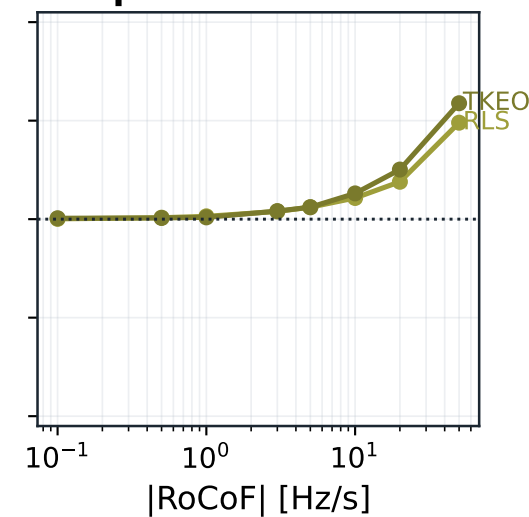
Window-based



Model-based



Adaptive



Data-driven

